

Rossby Wave Ray-tracing: A Modern Approach

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Abstract

1 Introduction

Rossby waves are large-scale waves that appear in stratified fluids on rotating bodies [2]. On earth, they appear both in the atmosphere and oceans and are almost perpendicular to the earth's axis of rotation. Atmospheric Rossby waves were first described in 1939 by Swedish-American meteorologist Carl-Gustaf Rossby.

This project will consider atmospheric Rossby waves and make use of modern numerical algorithms to analyze their behavior. Much of the work is based on the 1993 paper *Rossby Wave Propagation on a Realistic Longitudinally Varying Flow* by Hoskins and Ambrizzi [1]. This paper, however, differs in that different numerical methods for solving the problem will be compared.

2 Theory

This section serves as a brief introduction to the mathematical interpretations of Rossby waves, as well as to some of the numerical methods used.

2.1 The Stationary Wavenumber K_s

Central to any theory dealing with waves is the concept of a wavenumber K which describes the spatial frequency of a wave. Its definition is simply the reciprocal of the wavelength λ , i.e. $K = \frac{1}{\lambda}$. For

The central concept in the paper is the stationary wavenumber K_s . It tells us, at each location in the atmosphere, what wavelength a Rossby wave must have to remain stationary relative to the ground. The formula is:

$$K_s^2 = \frac{\beta_M - \partial^2 U / \partial y^2}{U} \quad (1)$$

Where U is the background zonal wind and β_M is the gradient of Earth's vorticity in Mercator coordinates.

When K_s is large and uniform over a region, that region acts as a waveguide: waves of the appropriate scale are trapped and can travel long distances. When K_s^2 becomes negative, waves cannot propagate at all (they are "evanescent").

2.2 Ray Tracing

Ray tracing treats a wave packet as a particle whose position (x, y) and wavenumber (k, l) evolve according to Hamilton's equations. The key idea is that waves bend toward regions of higher K_s , similar to how light bends in a medium with varying refractive index. By integrating these equations forward from a source location, we can predict where wave energy will go.

3 Method

The source code for this project is available in its entirety on github.

3.1 Data

The source data is provided by the European Centre for Medium-Range Weather Forecasts, and downloaded from Copernicus data store. It consists of winds in u - and v -directions at a pressure level of 300 hPa, analogous to the work of Hoskins and Ambrizzi. In the time domain, the data is provided monthly from January 1940 to May 2026. The spatial domain covers the entire Earth, and is supplied in a grid with size 0.25° .

Before any calculations are done, the entire dataset is time-averaged for the months December through February (DJF), for the entire available time period. Thus, the data used in the project is a 721×1440 spatial grid for the variables u and v . The code for time averaging the data is available in the file `timeavg.py`, and the final time-averaged data is available in the NumPy archive `djf_mean.npz` which contains variables `u`, `v`, `lat`, and `lon`.

3.2 Absolute vorticity

The absolute vorticity q depends on the curl of the wind field, and the Coriolis factor. Its definition is

$$q = \nabla \times (u, v) + 2\Omega \sin(\phi)$$

in which (u, v) is the wind vector field, Ω is the rotational frequency of the Earth, and ϕ is the latitude. Since the wind field is given in spherical coordinates, the (radial component of the-) curl operator $\nabla_R \times$ is defined as:

$$\frac{1}{R \cos \phi} \left[\frac{\partial v}{\partial \lambda} - \frac{\partial}{\partial \phi} u \cos \phi \right]$$

which simplifies to:

$$\frac{1}{R \cos \phi} \frac{\partial v}{\partial \lambda} - \frac{1}{R} \frac{\partial u}{\partial \phi} + \frac{u \tan \phi}{R}.$$

The derivatives are calculated using a central difference approximation, i.e. $f'(i) = \frac{f^{(i+1)} - f^{(i-1)}}{2h}$ with h being the grid size, and vice versa. For the boundaries at the poles, approximation is instead $f'(0) = \frac{f^{(1)} - f^{(0)}}{h}$. Since the dataset is cyclic longitudinally, the central difference approximation can be used as-is on the longitudinal 'boundary': $f'(0) = \frac{f^{(-1)} - f^{(1)}}{2h}$.

3.3 Computing K_s

3.4 Ray Tracing Implementation

4 Results

4.1 Comparison with Hoskins & Ambrizzi (1993)

5 Discussion

References

- [1] Brian J. Hoskins and Tercio Ambrizzi. Rossby wave propagation on a realistic longitudinally varying flow. *Journal of Atmospheric Sciences*, 50(12):1661–1671, 1993.
- [2] National Oceanic and Atmospheric Administration. What is a rossby wave?, 2024.